

DSO 498 – AI: Perils, Pitfalls and Promise

(A Previous version of this course was BUAD 498: AI: Seed for Change or Existential Threat?)

Spring 2022 – Tues 12pm – 1:50 pm (1hr 50 min/week) (Hybrid)
2.0 Units – Credit/ No-Credit Only – No Pre-Requisites – Upper Division Elective

Professor: Vishal Gupta

Contact: Use Course Slack!

Office Hours: *By appointment only. Ping me on Slack!*

Course Description

LinkedIn's 2020 Emerging Jobs Report identified AI Specialist as the #1 emerging job in the US, with 74% growth over the past 4 years, and CEOs across the world have publicly committed to investing in AI to transform their businesses. At the same time, leaders like Stephen Hawking and Elon Musk have described AI as an existential threat to humankind. So, which is it? Will AI usher in a new era of productivity, discovery, and global progress, or is it the first step towards an inevitable dystopian future?

In this course we will explore these and other questions surrounding artificial intelligence (AI) and machine learning in context of business and society. Through conversations with some of most respected thinkers in Artificial Intelligence across business, academia, and non-profits we will discuss

- What is Artificial Intelligence exactly? What can it do? What can't it do?
- What are some of the ways in which businesses and social enterprises are leveraging AI to effect change? What are the ingredients for success and the pitfalls to avoid?
- How does AI erode our privacy and safety? What are the ethical and moral implications of AI systems?
- How can we use AI to combat systemic racism, climate change and global poverty, without risking losing our own humanity to the cold-hearted logic of a machine?

Although this class focuses on guest speakers and group discussion, at the end of the day, answering these questions will fall to you and your peers as our next generation of leaders. You will be the ones responsible for developing AI and shaping it to as a tool for change, either for better or worse. What time better than now to start?

Learning Objectives

By the end of this course, you will be able to

1. Discuss modern themes in artificial intelligence (AI), machine learning (ML) and business analytics using domain-specific language and terms.
2. Identify opportunities across business domains and applications that could benefit from the integration of AI/ML.



3. Critically assess the ethical implications, practical challenges, and potential pitfalls of recent high-profile successes in AI/ML.

Required Materials and Course Information

There are no required textbooks for this class. However, I will distribute articles, podcasts, and videos to be reviewed before each class. These will be distributed via the class's Slack channel. Slack is available for free from USC ITS here: <https://itservices.usc.edu/slack/> See the same link for some tutorials if you are new.

Slack will also be the primary form of communication for the course. You *should* be posting regularly. (See below for expectations.) You can also contact me and the course TAs through Slack.

Prerequisites and/or Recommended Preparation:

This course has no pre-requisites.

Course Notes:

This course will be mostly conducted via synchronous Zoom sessions. Indeed, apart from a few lectures from me, the course will feature guest speakers who are experts across Artificial Intelligence (AI) and Machine Learning (ML) and their impacts on society, giving you a unique opportunity to learn from thought leaders in academia, industry, government, and non-profit spaces. To make it easier to draw the best-of-the-best to speak in the class, we have opted to make these sessions online via Zoom. For those of you on campus, there will be a dedicated classroom where you can use your laptop/tablet to sign-into class. (Please bring headphones to be considerate to others in the room who are also on the call.)

For the remaining sessions that I will be leading, these sessions will be in-person in the classroom and simultaneously be cast via Zoom. Hence, you can choose to attend these sessions in person or remotely. (Students in previous years have requested the in-person component.)

Regardless of whether you are attending remotely or in person, all sessions are SYNCHRONOUS. Students who cannot attend synchronously and have been granted accommodations should see below for asynchronous participation equivalents. Accommodations are granted by OSAS, and you should plan to schedule work commitments, interviews, and other obligations outside of class time. No exceptions. Plan ahead.

A Typical Session

For each session, we will have 5-10 (depending on class enrollment) session leaders. You can volunteer to be a session leader for the sessions that are most exciting for you. Session leaders play a special role in that session and it is an ideal way to develop a closer relationship with the speaker.

A typical session will be structured as follows:

- Before class, you will be assigned some short reading, podcasts, or videos to review. There will be a small number of reading comprehension questions on Blackboard for you to complete BEFORE class. Everyone must complete this step.
- At the beginning of class, I will debrief some of the reading, answer any pressing questions and help us connect the reading to the speaker for that session.
- We **may** divide up into breakout rooms to brainstorm questions for our guest speaker. Each breakout room will have one of the volunteer session leader that will help facilitate discussion. TA's and I will also ``drop in'' to these breakout rooms. By the end of the breakout session, the volunteer session leader will be responsible for uploading some questions for the speaker based on your discussion to a shared Google doc.
- We will reconvene as a group, and I will debrief the breakout room sessions by highlighting some of the common themes/questions that emerged across rooms.
- We will hear from our guest speaker, followed by an informal podcast style conversation with the speaker, me, and, the volunteer session leaders. This provides the session leader a unique opportunity to interact directly with the speaker.
- Time permitting, we will open things up to group discussion and questions from the rest of class.
- After class, you will post some reflections, comments, points for further discussion to our class Slack to “keep the conversation going” with your peers.

To recap, volunteer session leaders have a couple extra duties:

- They need to coordinate the discussion in a breakout room.
- They need to keep their camera on and remain engaged with the speaker during the presentation. Questions and conversation are highly encouraged! This is an opportunity for you to network and make a good impression.

Because of the extra duties, session leaders do not need to submit a post-class reflection for their specific sessions.

Use of Recordings

All sessions will be recorded and distributed via Slack to class members only. **Please do not share these recordings with anyone outside the class.** Pursuant to the USC Student Handbook (www.usc.edu/scampus, Part B, 11.12), students may not record a university class without the express permission of the instructor and announcement to the class. In addition, students may not distribute or use notes or recordings based on University classes or lectures without the express permission of the instructor for purposes other than personal or class-related group study by individuals registered for the class. This restriction on unauthorized use applies to all information that is distributed or displayed for use in relationship to the class.

Seriously. Many of these speakers are high-profile thought leaders that have agreed to spend their time with YOU, the class participants. They have not consented to sharing these talks with students outside the class, so please do not share them widely.

Synchronous Attendance Policies

1. Although this class is entirely online via Zoom, active class participation in breakout rooms and in-class exercises is crucial to achieving the learning objectives for this course. Hence,

unless students provide an accommodation letter from USC OSAS or from Marshall detailing visa or travel restrictions, **“synchronous” attendance in Zoom sessions is required.**

2. Any student with an OSAS or Marshall accommodation that prevents them from attending class synchronously should submit documentation to me within the first two weeks of class. Such students should plan to watch the recording of the class and complete the “asynchronous participation equivalent” described below for each session missed.
3. Medical emergencies: In rare occasions, you may fall ill. Students who are experiencing illness and cannot attend class synchronously should contact the teaching team via Slack and focus on getting well. Once healthy, we can discuss a timeline for which students can complete “asynchronous participation equivalents” for the missed session/sessions.

Asynchronous Participation Equivalent

1. As described above, there may be instances where you will need to miss a class session and have been granted an accommodation to do so by me. In such cases, you are still responsible for all material covered, including watching the posted recording of the session.
2. If you miss a synchronous session due to an OSAS approved accommodation, you should complete any pre- or post-class assignments for the session and submit them by the same deadline as posted for your classmates on Slack or in this syllabus.
3. If you miss a synchronous session due to an unexpected illness (“Medical emergency” above), you should coordinate with the teaching team as soon as possible to agree on a timeline when you will complete these assignments.
4. In *addition* to the above, you should plan to post a 150 - 200 word reflection on Slack connecting the pre-class reading materials to the guest speaker’s talk. Your reflection should also raise some discussion points to spur further conversation on Slack – much as you would have had you participated in the in-class breakout room. Again, for clarity, this reflection is in *addition* to any pre- and post-class assignments for that session, so for a missed session you may need to answer 1 set of pre-class questions on Blackboard AND ALSO post 2 reflection pieces.

Grading Policies:

This course will be graded as Credit/No Credit. Your grade is computed by the following breakdown:

Pre-Class Assignments	20%
Synchronous Participation/ Asynchronous Participation Equivalent	20%
Post Class Discussion on Slack	20%
Final Team Presentation	40%

There are no exams but there is a Final Team Project. You must complete *ALL* assignments above (late if you must) in order to pass the course.

See below for some more details:

Pre-Class Assignments

As mentioned, I will post a collection of readings/podcasts/videos to Slack before each session to inspire you. I will also post some short comprehension questions on this material to Blackboard. These questions are not meant to be onerous, but rather just to help keep you “on track.” If you’ve completed the pre-class material, answering the questions should be very easy. All Pre-Class Assignments are due before the start of their respective session. No exceptions.

Synchronous Participation / Asynchronous Participation

Your participation will be assessed primarily on the quality and thoughtfulness of your contributions. For those participating in a session synchronously, you will be assessed on

- Breakout room discussions – TAs and I will randomly drop into some breakout rooms to listen to discussion and help guide you forward. We may visit you multiple times in a semester, or may not visit you at all (such is the nature of randomness). Regardless of whether a member of the teaching team is present, strive for high-quality, thoughtful, engaged discussion! This entails having your camera on, speaking up, and conversing with your classmates. Breakout rooms will be recorded, and the teaching team may review recordings after a session.
- (Session Leaders Only) Questions for Speakers – Session leaders are responsible for adding 2-3 questions that emerge from your breakout discussion for the speaker to a shared Google doc. Strive for thoughtful questions that probe the issues you care about most! Creative questions that connect with broader themes in the course are most valued.

Your grade for participation will be a combination of any observation sessions from me and the TAs (either in real-time or viewing recorded breakout sessions) and by evaluating the quality of the questions your team adds to the shared Google doc.

For those participating asynchronously, the above two assessments will be replaced by your asynchronous participation equivalent assignment described above. Asynchronous participation equivalents are due 48 hours after a session concludes unless you have made other arrangements with the teaching team due to illness.

Post-Class Discussion on Slack

Based on the speaker’s lecture, I will post some discussion questions to Slack after class. Consider these questions to be *starting points* for a conversation. You should post a 150-200 word reflection on either my discussion question, a previous response from a classmate (ideally by replying in a thread), or a new element of the speaker’s talk that you to further explore as a group. **All post-class discussion contributions are due by 11:59 on Sunday following a session.**

Final Presentation

Obviously, not every aspect of the rapidly evolving role of AI/ML in our world can be covered in class. The final presentation is an opportunity for you to work in a small team (4-5 students) to explore a contemporary issue that we have yet to discuss in-depth in class. As a team, you will research this topic and create a brief presentation that reviews essential background information, poses some critical questions we as a society must examine, and discusses the implications for your classmates – Marshall undergraduates preparing to enter a data-driven world.

You are free to form your own teams of 4-5 students (details to be announced on Slack). Students who do not choose to form or join a team by **1 April 11:59 pm** will be assigned to one.

Presentations must include 5 slides (NOT including a title slide with students' names), and the presentation should be approximately 5 minutes long. A PDF file of the team presentation slides and a link to the recording should be uploaded to Blackboard by 4 May 11:59 pm. Note: teams are free to use their preferred software to record their presentation, however, Zoom may be the easiest way for most groups. By selecting to save your recording to the cloud, Zoom will generate a URL link to the recording automatically, which you can submit with the pdf file. For those teams who do not use Zoom, you must submit an mp4 recording of the presentation. Additional information about this assignment will be posted on Slack.

You will be asked to complete a peer-evaluation regarding the final project. Your grade will be assigned by me and the course TAs after reviewing your presentations and these peer-evaluations.

Finally, and most importantly, to celebrate your work, we will be voting on some of the best presentations to screen and discuss in the final exam session. This voting will not affect your grade for the final project or the course. (Details to follow on Slack.)

Assignment Submission Policy:

All assignments for the course must be turned in the due date/time electronically, including in-class assignments. Depending on the assignment, this may involve

- Adding to a shared google document the class
- Posting to our class's Slack discussion channels
- Uploading a document to Blackboard

Check the assignment for details on submission.

Late submissions (even by a few minutes) will be penalized. Late or not, you **MUST** submit all assignments for the course (or asynchronous equivalents) to pass the course.

Zoom Etiquette Expectations:

It perhaps goes without saying, but many of the speakers joining us this semester are incredibly well-respected in the field and have graciously offered their time to speak to you. I expect you to show them the respect and professionalism they deserve. It's up to you be ambassadors for the Trojan Family.

Consequently, I will have some serious expectations for Zoom Etiquette throughout EVERY session.

- Strive to keep your camera on all the time. If you need to hide your video for a moment to step away, that's ok, but only a moment. Speakers should not be speaking to a faceless audience.
- Everyone should be in an environment where you can easily unmute and participate in class if called upon, or if you have a question. This means BOTH being somewhere quiet and private, but ALSO somewhere with good internet connectivity.
- Cameras are on in Breakout Rooms and you should participate actively.
- You should not be multitasking during class. If you want to take notes, great. But if I get the sense, you are not following the discussion, I **will** cold-call you.
- **For Session Leaders: YOU MUST KEEP YOUR CAMERA ON and be engaged throughout the entire session (100% of the time!).** You're going to serve as the primary point of contact for the speaker to USC. Represent us well!

If for some reason you're not meeting these standards, I will give you one and only one polite warning. If you continue to disregard the standards, you will not receive synchronous participation credit for the session.

COURSE CALENDAR -- Subject to change

I am still in the process of scheduling speakers for next semester. Below is the schedule from last year to give an indication of the themes and kinds of speakers. Check back soon for updates!

<u>Date</u>	<u>Topic and Speaker</u>	<u>After Class</u>
11 Jan	"Talk the Talk" Vishal Gupta, PhD USC Marshall, Data Sciences and Operations - Briefly review syllabus/course logistics - Landscape of AI, ML and Analytics - Defining Key terms	Submit post-class Slack Discussion contribution by 15 Jan 2022 at 11:59 pm.
18 Jan	What makes an AI/ML tool "good"? Vishal Gupta, PhD USC Marshall, Data Sciences and Operations	Submit post-class Slack Discussion contribution by 22 Jan 2022 at 11:59 pm.
25 Jan	Topic: AI in Finance Title: Artificial Intelligence in Algorithmic Trading and Finance Iain Dunning, PhD Lead, Hudson River Trading AI Labs Former Research Lead at DeepMind	Submit post-class Slack Discussion contribution by 29 Jan 2022 at 11:59 pm.
1 Feb	Topic: AI for Social Good Title: Using AI to Combat Sex-Trafficking on the Deep Web Emily Wyatt Director, Counter Human Trafficking Initiatives at Uncharted Software Co-Chair of Tellfinder Alliance & Pia Ramchandani Director of Responsible AI, PricewaterhouseCoopers	Submit post-class Slack Discussion contribution by 5 Feb 2022 at 11:59 pm.
8 Feb	Topic: AI in High-Stakes Precision Medicine Title: Dissecting Algorithmic Bias Ziad Obermeyer, MD Blue Cross of California Distinguished Professor of Health Policy and Management at the Berkeley School of Public Health	Submit post-class Slack Discussion contribution by 12 Feb 2022 at 11:59 pm.
15 Feb	Topic: Designing a Data Science Product Title: Building and Scaling ML Products for the Real World	Submit post-class Slack Discussion contribution by 19 Feb 2022 at 11:59 pm.



	Justin Norman Vice President, Data Science at Yelp	
22 Feb	Topic: Intellectual Property, Legislation, and the Future of AI Title: Regulating Algorithms: Law, Policy, and Governance Christopher Bavitz Wilmer Hale Clinical Professor of Law Managing Director, Cyberlaw Clinic at the Berkman Klein Center for Internet and Society Vice Dean for Experiential and Clinical Education	Submit post-class Slack Discussion contribution by 26 Feb 2022 at 11:59 pm.
1 Mar	Topic: Privacy, Security and Vulnerabilities Title: Our Devices Scream Our Passwords! Samy Kamkar Security Expert, Whistleblower, Whitehat Hacker Chief Security Officer OpenPath Security, Inc.	Submit post-class Slack Discussion contribution by 5 Mar 2022 at 11:59 pm.
8 Mar	Topic: Algorithmic Bias Title: Quantifying Bias in Human and Machine Decisions Sharad Goel, PhD Professor of Public Policy at Harvard Kennedy School of Government Founding Director of the Computational Policy Lab at Stanford, University Member of the Open-Policing Project	Submit post-class Slack Discussion contribution by 12 Mar 2022 at 11:59 pm. No Post-Class This Week
15 Mar	SPRING BREAK	
22 Mar	Topic: Reinforcement Learning Basics Speaker: Vishal Gupta	Submit post-class Slack Discussion contribution by 26 Mar 2022 at 11:59 pm. 27 Mar 2022 at 11:59 pm
29 Mar	Topic: Using AI to Fight Climate Change (I) Title: Tracking Real time Atmospheric Carbon Emissions Speaker: Aaron Davitt Principal Analyst, Remote Sensing Wattime.org	Choose Project teams by 1 April 11:59 pm Submit post-class Slack Discussion contribution by 3 April 2022 at 11:59 pm
5 Apr	Topic: AI and Sports Analytics Tentative Title: "The Evolving world of Sports Analytics" Kevin Meer NFL Analyst at Zelus Analytics	Submit post-class Slack Discussion contribution by 9 Apr 2022 at 11:59 pm. 10 Apr 2022 at 11:59 pm.
12 Apr	Topic: Using AI to Fight Climate Change (II) Title: TBD Sam Gill Co-Founder and COO Sylvera	Submit post-class Slack Discussion contribution by 16 Apr 2022 at 11:59 pm 17 Apr 2022 at 11:59 pm

19 Apr	Topic: AI and Investigative Journalism Title: TBD Chery Phillips Director, Big Local News Hearst Professional in Residence, Graduate Program in Journalism, Stanford University Director, Stanford Computational Policy Lab	Submit post-class Slack Discussion contribution by 23 Apr 2022 at 11:59 pm 23 Apr 2022 at 11:59 pm
26 Apr	Topic: Course Wrap-up Speaker: Vishal Gupta, PhD USC Marshall, Data Sciences and Operations	No Post-class reflection!! But please do your Course Evaluations!
<u>Final Project Screening</u> <u>11 May 2-4 pm</u>	Viewing of selected Final Project Presentations! (Attendance is required by University)	Final Project Due on BB by 4 May 11:59 pm.

ADDITIONAL INFORMATION

Use of Recordings

Pursuant to the USC Student Handbook (www.usc.edu/scampus, Part B, 11.12), students may not record a university class without the express permission of the instructor and announcement to the class. In addition, students may not distribute or use notes or recordings based on University classes or lectures without the express permission of the instructor for purposes other than personal or class-related group study by individuals registered for the class. This restriction on unauthorized use applies to all information that is distributed or displayed for use in relationship to the class.

USC Statement on Academic Conduct and Student Well Being Support Systems

Academic Conduct:

Plagiarism – presenting someone else’s ideas as your own, either verbatim or recast in your own words – is a serious academic offense with serious consequences. Please familiarize yourself with the discussion of plagiarism in SCampus in Part B, Section 11, “Behavior Violating University Standards” policy.usc.edu/scampus-part-b. Other forms of academic dishonesty are equally unacceptable. See additional information in SCampus and university policies on [Research and Scholarship Misconduct](#).

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University’s educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Support Systems:

Counseling and Mental Health - (213) 740-9355 – 24/7 on call
studenthealth.usc.edu/counseling

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

National Suicide Prevention Lifeline - 1 (800) 273-8255 – 24/7 on call
suicidepreventionlifeline.org

Free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week.

Relationship and Sexual Violence Prevention Services (RSVP) - (213) 740-9355(WELL), press “0” after hours – 24/7 on call
studenthealth.usc.edu/sexual-assault



Free and confidential therapy services, workshops, and training for situations related to gender-based harm.

Office for Equity, Equal Opportunity, and Title IX (EEO-TIX) - (213) 740-5086
eeotix.usc.edu

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

Reporting Incidents of Bias or Harassment - (213) 740-5086 or (213) 821-8298
usc-advocate.symlicity.com/care_report

Avenue to report incidents of bias, hate crimes, and microaggressions to the Office for Equity, Equal Opportunity, and Title for appropriate investigation, supportive measures, and response.

The Office of Student Accessibility Services (OSAS) - (213) 740-0776
osas.usc.edu

OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

USC Campus Support and Intervention - (213) 821-4710
campussupport.usc.edu

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

Diversity, Equity and Inclusion - (213) 740-2101
diversity.usc.edu

Information on events, programs and training, the Provost's Diversity and Inclusion Council, Diversity Liaisons for each academic school, chronology, participation, and various resources for students.

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call
dps.usc.edu, emergency.usc.edu

Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety - UPC: (213) 740-6000, HSC: (323) 442-120 – 24/7 on call
dps.usc.edu

Non-emergency assistance or information.

Office of the Ombuds - (213) 821-9556 (UPC) / (323-442-0382 (HSC)
ombuds.usc.edu

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-3340 or otfp@med.usc.edu
chan.usc.edu/otfp

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.